

%>% The pipe operator |>

Day 5 - Friday, May 22, 2026

i Today: Friday May 22

Before class

- Read *Data Computing* §3.6 ([Click here](#)).

Plan for today

- Quiz 1 (20 min)
- The pipe operator
- Chaining commands together
- In-class activity

Helpful links

- [Syllabus](#) · [AI policy](#) · [Schedule](#) · [Data Computing §3.6](#)

Quiz 1

Quiz 1 — instructions

! Closed-book, no AI, no electronics

- Put away phones and laptops.
- You may use a pen or pencil.
- 20 minutes.
- I'll hand out the quiz now and collect it at the end.

Topics covered:

- R as a calculator and basic syntax
- Data types and structures
- Objects and assignment
- Reading R code (functions, arguments, variables, constants)
- Packages: installing vs. loading
- Histograms in `ggplot2` (reading code, not writing from scratch)

When you finish early, flip your quiz face-down and wait quietly.

Quiz debrief

A few quick notes:

- We'll go over common stumbling points on Monday after I've graded them.
- If you finished with time to spare: nice.
- If you didn't finish: don't panic. Quizzes are 15% of the grade, weighted across the semester.

Now let's get into something new.

The pipe operator

A problem we've already seen

Remember this slide from Wednesday?

```
sqrt(mean(c(1, 4, 9, 16, 25)))
```

```
[1] 3.316625
```

To read this, your eye has to jump **inside-out**:

1. First `c(1, 4, 9, 16, 25)` makes a vector.
2. Then `mean(...)` of that vector is 11.
3. Then `sqrt(11)` is 3.317.

That's already tough. Imagine four or five nested functions.

A real example

Suppose we want to:

1. Take the `penguins` data,
2. Keep only rows where `body_mass_g` is not missing,
3. Group by `species`,
4. Compute the mean body mass for each species.

Written with nested function calls:

```
summarize(  
  group_by(  
    filter(penguins, !is.na(body_mass_g)),  
    species  
  ),  
  avg_mass = mean(body_mass_g)  
)
```

You have to read it from the **inside out**, and the arguments get separated from their functions. Ugh.

Enter the pipe

The **pipe operator** lets you read code left-to-right, top-to-bottom — the way you actually think about it.

```
penguins |>  
  filter(!is.na(body_mass_g)) |>  
  group_by(species) |>  
  summarize(avg_mass = mean(body_mass_g))
```

Read this as:

“Take `penguins`, **then** filter out missing values, **then** group by species, **then** summarize the average mass.”

That’s it. That’s the whole idea.

What the pipe does

```
x |> f()
```

is **exactly the same** as

```
f(x)
```

The pipe takes whatever's on the **left** and inserts it as the **first argument** to the function on the **right**.

```
# These do the same thing:  
sqrt(16)
```

```
[1] 4
```

```
16 |> sqrt()
```

```
[1] 4
```

With multiple arguments

```
x |> f(arg2, arg3)
```

is the same as

```
f(x, arg2, arg3)
```

The pipe always fills the **first** argument; the rest go inside the parentheses as usual.

```
# Without the pipe:  
round(3.14159, digits = 2)
```

```
[1] 3.14
```

```
# With the pipe:  
3.14159 |> round(digits = 2)
```

```
[1] 3.14
```

Chaining pipes

The real power: you can chain pipes together to do many steps in order.

```
c(1, 4, 9, 16, 25) |>
  mean() |>
  sqrt()
```

```
[1] 3.316625
```

Compare to the nested version:

```
sqrt(mean(c(1, 4, 9, 16, 25)))
```

```
[1] 3.316625
```

Same answer. The piped version reads in the order you do the work.

Style: one pipe per line



Convention

Put each pipe step on its **own line**, with the `|>` at the **end** of the previous line. Indent the continuation by 2 spaces.

Good:

```
penguins |>
  filter(!is.na(body_mass_g)) |>
  group_by(species) |>
  summarize(avg_mass = mean(body_mass_g))
```

Not so good (works, but harder to read):

```
penguins |> filter(!is.na(body_mass_g)) |> group_by(species) |> summarize(avg_mass = mean(body_mass_g))
```

Two pipes?

`|>` vs. `%>%`

You will see **two** pipe operators in R code online:

Table 1: The two pipes in R.

Pipe	Where it comes from	When introduced
<code>%>%</code>	The <code>magrittr</code> package (and tidyverse)	2014
<code> ></code>	Base R	2021

They do **almost the same thing**. For everything we'll cover in this course, they behave identically.

Why two pipes?

- `%>%` came first. It's the original tidyverse pipe.
- Base R added `|>` later because the pipe became so popular they wanted it built in.
- The textbook (*Data Computing*) uses `%>%`.
- I'll use `|>` in lecture going forward — it's the modern default.

Note

You can use either one. I'll accept both on assignments. Just be consistent within a single document.

Typing the pipe quickly

You don't have to type `|>` character-by-character. RStudio has a shortcut:

- **Mac:** `Cmd + Shift + M`
- **Windows/Linux:** `Ctrl + Shift + M`

Tip

By default, this shortcut inserts `%>%`. To make it insert `|>` instead:
Tools > Global Options... > Code > Use native pipe operator

Reading piped code

Translating between styles

You'll see code in both styles all over the internet. Practice translating in your head.

Nested:

```
head(arrange(filter(penguins, species == "Gentoo"), body_mass_g), 5)
```

Piped:

```
penguins |>
  filter(species == "Gentoo") |>
  arrange(body_mass_g) |>
  head(5)
```

Same result. **Way** easier to read the piped version.

What the pipe is NOT

⚠ The pipe doesn't run things in parallel

It just rewrites your code. Each step still runs one at a time, in order.

⚠ The pipe is NOT the same as + in ggplot2

- dplyr and other data verbs: use |> or %>%.
- ggplot2 layers: use +.

This trips everyone up. Keep them straight.

```
# Correct:
penguins |>
  filter(!is.na(body_mass_g)) |>
  ggplot(aes(x = body_mass_g)) +
  geom_histogram()
```

A peek at what's coming

You'll see the pipe everywhere from here on out. Tomorrow we start the **dplyr** verbs:

- `filter()` — pick rows
- `select()` — pick columns
- `arrange()` — sort
- `mutate()` — make new columns
- `summarize()` — collapse to a single number
- `group_by()` — split into groups before summarizing

Every one of these is designed to be chained together with the pipe.

In-class activity

Activity: pipe practice (~ 15 min)

Work individually or with a neighbor.

Setup:

1. Open RStudio.
2. New R script. Save as `day5-activity.R`.
3. At the top:

```
library(tidyverse)
library(palmerpenguins)
```

Goal: Translate between nested and piped code, then write your own.

Activity: Part 1 — translate to piped

Rewrite each of these using the pipe `|>`. Don't run them yet — just rewrite the code.

```
# 1
sqrt(64)

# 2
mean(c(2, 4, 6, 8, 10))

# 3
```



```
round(sqrt(50), digits = 2)

# 4
head(arrange(penguins, body_mass_g), 3)
```

Then run both the original and your piped version. Confirm you get the same answer.

Activity: Part 2 — translate to nested

Rewrite each of these **without** the pipe. Then run both versions.

```
# 5
c(1, 2, 3, 4, 5) |> sum()

# 6
penguins |> head(2)

# 7
9 |> sqrt() |> round(1)
```

Activity: Part 3 — write your own

Write a single piped expression that:

1. Takes `penguins`,
2. Keeps only the rows where `species` is "Adelie",
3. Picks just the first 5 rows.

Hint

You'll need `filter()` and `head()`. Chain them with `|>`.

To turn in: Save your script. We'll check at the end of class.

Activity debrief

Common things to watch for:

- **The pipe goes at the END of the line**, not the start. `x |>` then a new line — not `\n |> f()`.

- The first argument is filled automatically. Don't write `penguins |> filter(penguins, ...)` — drop the first `penguins`.
- `+` vs. `|>` again. `dplyr` verbs chain with `|>`; `ggplot` layers chain with `+`.
- Parentheses still matter. `f()` not `f`. The pipe doesn't change that.

Wrap-up: what we covered today

- The pipe operator (`|>` or `%>%`) takes the value on the left and feeds it as the first argument to the function on the right.
- Pipes let you read code top-to-bottom in the order you'd actually do the work.
- Each step gets its own line; the pipe lives at the **end** of the previous line.
- `|>` and `%>%` are nearly interchangeable for our purposes.
- `dplyr` verbs chain with `|>`; `ggplot` layers chain with `+`.

Wrap-up & looking ahead

Monday: `dplyr` verbs — `filter()`, `select()`, `arrange()`.

Upcoming Deadlines

- Today, 11:59 p.m. — [HW 1: Basic R & Plots](#)
- Mon 5/25, before class — DC §7 reading (Canvas link TBA)

Reach out

Stuck? Office hours, or email me (cgc5478@psu.edu) or Xinyue (xpw5228@psu.edu).

Reference

[Syllabus](#) · [AI policy](#) · [Schedule](#) · *[Data Computing](#)*